

# The inflationary outcome of a monetary policy shock: will the ECB stabilise inflation in the aftermath of the 2022 contraction?

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## Abstract

We investigate the effect of a contractionary monetary policy shock on price level in the Euro area, in order to understand if the tightening policy conducted by the ECB will be able to stabilize inflation. The monetary shock is identified along with a spread and an information shock following [Fanelli and Marsi(2022)], then the IRF is estimated by means of an Instrumental Variable Local Projection à la [Stock and Watson(2018)] with "standard" confidence intervals and using the monetary policy factor as an instrument. Our results do not bring any statistical evidence supporting the effectiveness of the monetary policy shock in stabilizing the surge in inflation recently experienced by the Euro Area. We also observe that to assess the effectiveness of monetary policy, the supply or demand nature of inflation needs to be taken into account, alongside with the possible de-anchoring of inflation expectations.

## 1 Introduction:

How relevant is **monetary policy** nowadays? What impact does it exert on inflation? How can we perform this assessment using macroeconometrics methods? Central bankers and academics have been asking these critical questions for decades. Never before in the history of fiat money as much as today there has been such a broad consensus on the benefits of a low-inflation environment. The surge of prices experienced in the **Euro Area(EA)** is the inevitable market response to the breakdown of globalization. The pandemic-lead disruption of supply chains, the Ukrainian War, and the related implications of European sanctions and ultimately the Israeli-prompted hostilities in the Middle East all play a role in disrupting European markets. To counteract upward inflationary movements and the threats of uncertainty, central is the tool of monetary policy which, according to Macroeconomic theory<sup>1</sup>, affects the inflation level through the effects on **aggregate demand**. Accordingly, crucial is the role of the European Central Bank. The institution contributes to its objective of price stability by managing monetary policy directly through its conventional and unconventional policy instruments and through communication. Indeed, information signaling is as relevant as policy in terms of its impact on markets. But how consistent is the EA price level response to monetary policy? Are the ECB's efforts enough to achieve its target inflation? Drawing on previous contributions, especially on that of [Fanelli and Marsi(2022)], our research explores the questions, contributing to literature through the application of a novel method, the **Local Projection (LP)** methodology of [Jordà(2005)], to the estimation of the impact of an *unconventional monetary policy shock on inflation*. First, in a replication of [Fanelli and Marsi(2022)], ECB's unconventional **monetary policy shocks** are studied and their identification is achieved. Then, we employ the novel methodology of Local Projections [Jordà(2005)] to model their transmission. Here a special focus is dedicated to assess the impact of monetary policy on inflation. Ultimately,

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<sup>1</sup>[Galí(2015)]

empirical evidence and theoretical arguments are proposed to envisage whether the ECB’s recent monetary policy interventions undertaken in the Summer of 2022 will likely achieve a renewed stability.

## 1.1 The monetary policy shock:

Generally speaking, a shock is a primitive exogenous force that is uncorrelated with other shock and is economically meaningful. The identification of a monetary shock is particularly difficult as monetary policy is for its nature very **endogenous**. As we are interested in an exogenous shock, its identification is challenging as a lot of confounding factors can negatively affect the estimates. If you were interested in the ability of the central bank to affect output you may be tempted to regress output change on interest rate change, but the central bank intervention could be induced by a financial crisis (ex. 2008) that is a **confounding factor** that affects output and so is in the error term. This is what we mean by endogeneity of the monetary policy: every movement of interest rates that is observed is a response to something; but what is not a response is appointed as a monetary shock. The fact is that you can’t just determine the effects of monetary policy by looking at interest rate changes; most of the changes in interest rates are endogenous and much bigger than the changes caused by the monetary policy shock alone. In fact, interest rates continuously change and are only partially controlled by central banks, making them an imperfect indicator of monetary policy. [Christiano and Evans(1999)] introduce monetary shocks from the following expression

$$S_t = f(\Omega_t) + \sigma_t \epsilon_t \quad (1)$$

Where  $S_t$  is a measure of monetary policy activity (say the funds rate),  $f(\Omega_t)$  is the systemic response of the monetary policy given the information set  $\Omega_t$  and the random variable  $\sigma_t \epsilon_t$  is the monetary policy shock and  $\sigma_t$  its standard deviation. In this sense some examples of monetary shocks are time variation in policymakers’ preferences and goals or time variation in policymakers’ beliefs about how the economy works, political influences or pursuit of other objectives such as exchange rate stability. All these examples in fact are **exogenous shocks** and do not depend on the central bank endogenous reaction to macroeconomic variables.

## 1.2 Literature overview

As regards the issue of **identification** of a monetary policy shock, a relevant contribution is that of [Romer and Romer(2004)]. The authors exploit the “Greenbook” forecasts, to regress fed funds rate changes on Federal Reserve’s internal forecasts (endogenous and systemic response) and to treat the residuals as a measure of exogenous variation in the funds rate. A different approach is that proposed in the framework of the High Frequency Identification (HFI) method. Conceptually, this methodology is based on the idea that if changes in interest rates surprises (fed funds future for example) in an enough tight amount of time around the FOMC meetings are dominated by the monetary shock; this is because other confounding factors that affects interest rates supposedly remain fixed. Of course you need high-frequency data (you cannot use a similar argument to identify effects on output, as you don’t observe it enough closely and different confounding factors are very likely to affect it). Some threats to identification consist in the fact that you cannot make the interval the smallest possible as financial markets can take some time to react, hence the optimal window is thought to be a **30-minute window**. This kind of approach was first used by [Cook and Hahn(1989)], [Kuttner(2001)], [Cochrane et al.(2002)], and consists in using changes in interest rates and asset prices surprises in order to identify the monetary shock. A more recent literature started by [Nakamura and Steinsson(2018)] found that the nature of surprises around FOMC announcements are also to be attributed to an information effect (**information shock**). Information disclosure by central banks are thought to reveal high quality information about the economy and so to affect private sector’s expectations leading to changes in interest rates; ignoring this shock would cause biased estimation of the monetary shock. Fanelli and Marsi enrich the literature on identification of monetary shocks by proposing a three-shocks identification framework. They pose that not only a monetary policy shock and an information shock represent EA unconventional monetary policy

surprises. If we consider the Euro area another shock is presented to take the scene. Their argument is that the ECB is likely to be concerned about **fragmentation** risk and so it affects default's expectations; hence monetary policy announcements are also for another shock, the "**spread shock**". Overall, [Fanelli and Marsi(2022)]'s newly identified shocks are the following:

- **Monetary policy shock:** when it is contractionary, it increases interest rates by raising the default free yield curve; it lowers the stock index and increases the periphery risk premium.
- **Information shock:** in the case it causes a positive revision of expectations about the economy, it likely increases the interest rate, the stock index and lowers the periphery risk premium.
- **Spread shock:** it is related to an opposite movement of default-free long term rates and periphery risk premium.

Note that from a qualitative point of view the spread and the information shock are very similar, as they are both about information disclosure. This requires to exploit some distinguishing features through the usage of a narrative restriction. Our work follows [Fanelli and Marsi(2022)], who first used factor analysis to extract three factors from [Altavilla et al.(2019)] database as a measure of the three shocks (see [Gürkaynak et al.(2004)]). Then, they used them as instruments in the computation of a proxy-SVAR and its IRFs à la [Angelini and Fanelli(2019)]. Another work that is related to [Fanelli and Marsi(2022)] is the one of [Motto and Özen(2022)] and both are connected to [Altavilla et al.(2019)] who applied [Gürkaynak et al.(2004)] methodology of factor extraction in the EA. By considering the presence of three shocks in the EA, we intend to identify a monetary policy shock that represents the effective intervention of the ECB and that is not biased by other contemporaneous shocks or confounding factors. A further step is the building of an **impulse response function (IRF)** with the previously identified monetary policy. This task is accomplished by means of a **Local Projections** method [Jordà(2005)]. As regards contributions in literature to the methodology of Local Projections, the effect of monetary shocks on inflation through LP methods has been investigated by [Murgia(2020)] in the EA and [Champagne et al.(2018)] in Canada. Results from these studies are not very consistent as regards their assessment of the LP method. This is coherent with the nature of local projections, that although are very flexible, discount this flexibility in terms of precision.

## 2 Data:

Throughout the paper, the **Euro-Area Monetary Policy event study dataset (EA-MPD)** constructed by [Altavilla et al.(2019)] is employed. The timeframe spanned by the database is January 2002 to June 2020 <sup>2</sup>. The database is a long -for euro standards- time series of policy-dates event studies for the euro area, chosen to measure and assess the monetary policies adopted by the European Central Bank (see appendix A 6). The EA-MPD features price changes for a broad class of assets and various maturities, including Overnight Index Swaps (OIS), sovereign yields, stock prices and exchange rates ,and is used in the process of factors extraction. Those data are collected around press release, press conference and monetary events. Indeed, monetary policy surprises are multi-dimensional <sup>3</sup> and are revealed in a multi-step procedure. It is now the standard for central banks to announce their monetary policy decisions immediately after policy meetings in the *press release*. Subsequently, these decisions are elaborated to bring greater transparency to the monetary policy decision-making process via *press conference*. This occasion constitutes a key vehicle of ECB communication. The ECB's press conference typically consists in an introductory statement by the President, followed by a Q&A session for journalists to gain clarification on monetary policy decisions. For during the event unconventional monetary policies are usually announced <sup>4</sup> and the related measures discussed, the ECB press conference represents a critical time impacting market movements. As explained

<sup>2</sup>In the daily proxy-SVAR the sample starts from August 2005, due to lack of data availability for the earlier period.

<sup>3</sup>Altavilla et al. find Target, Timing, Forward Guidance and Quantitative Easing surprises.

<sup>4</sup>A conventional monetary policy regime is one where the central bank sets the short-term interest rate as its policy instrument in response to macroeconomic shocks associated with inflation and the output gap. Deviations from this interest-rate rule generate a conventional monetary policy shock. In the unconventional policy regime, the central bank moves further, expanding its balance sheet under the zero lower bound. Unconventional policies comprise 'quantitative easing' and 'credit easing', depending on whether the central bank focuses on the liability side or the asset side of its balance sheet.

by [Fanelli and Marsi(2022)], the two-steps announcement procedure can be exploited to disentangle HF surprises of interest rates and asset prices into different components related to the different aspects of the policies pursued by the ECB. As our interest lies in unconventional monetary policies only, we focus on HF surprises observed around press conferences alone. Throughout our identification procedure, we assume that the data generating process—approximated by a Proxy-SVAR—operates at the *daily* frequency. Thus, in implementing a Proxy-SVAR analysis we use a collection of daily endogenous variables including: ecb 5 year yield, ecb 1 year yield, the 10 year spread IT-OIS and the log of the Euro stockxx50. Finally, in our last phase of construction of a local projection of the monetary shock to study inflation, we transform the data from daily to monthly. As macroeconomic variables such as inflation and industrial production are only observed on a monthly frequency, as shown by [Forni and Marcellino(2016)], we wouldn't be able to explore the effect of our identified *monetary policy shock* on *inflation* as this too is not available at daily frequency. To reduce the risk of losing relevant information, we employ a strategy recently discussed in [Gertler and Karadi(2015)]: we convert the three proxies (monetary, information and spread) from the daily to the monthly frequency through averaging. Finally, we quantify the dynamic causal effects produced by the *monetary policy shock* on *inflation* through an IV LP procedure.

### 3 Methodology:

#### 3.1 Identification:

Monetary shocks affect the economy not only directly but also through information signaling. Observable policy actions together with institutional communication channels transmit information from the policy maker to market participants. For instance, interest rate decisions transmit information about the central bank's assessment of the economic outlook [Hubert and Maule(2021)]. The transmission of monetary impulses can be strongly affected by the disclosure of information. The very ability of a central bank to stabilise the economy may result affected. Empirically, both price and output puzzles could occur. Due to asymmetric information, an agent could interpret a policy rate hike either as a deviation of the central bank from its monetary policy rule, i.e. a contractionary monetary shock, or as an endogenous response to inflationary pressures expected to hit the economy in the near future. Both the scenarios would cause a visible rate increase, but each of them imply profoundly different evolutions for macroeconomic aggregates, and related agents' expectations [Campbell et al.(2012)]. In addition, the announcement of credible growth-enhancing measures and prudent fiscal measures by a government following a monetary policy tightening could create positive expectations on the part of investors about the safety of their investments. This would reduce the perception of risk around that government's bonds, and therefore the spread. This would also affect agents' expectations and ultimately the evolution of macroeconomic aggregates. Fanelli and Marsi consider the problem of *asymmetry of information* between the central bank and market agents in the context of the Euro Area and its contribution to affect the market's and agents' expectations. Apart from the unconventional monetary policy channel directly affecting the economy, they consider information in the form of pure signaling by the central bank and fragmentation risk. They call the former the *information shock*. This shock, stemming from monetary policy announcements, has to do with the idea that during press conferences central banks, besides announcing monetary policy measures, also reveal to the public their view about the future state of the economy, thus influencing expectations of markets' participants [Fanelli and Marsi(2022)]. Further, they recognize the role of *fragmentation* in the transmission of monetary policy. This refers to interest rates in different countries in the monetary union reacting differently when the ECB revises its interest rate. For instance, when markets are most worried about the stability of Italy's government, the Italian-German spread likely widens. To account for this additional risk, Fanelli and Marsi introduce another component of the monetary shock, a **spread shock**, representing a monetary surprise that consist in the widening of sovereign spreads of peripheral countries as a result of ECB's announcements. This shock is relevant in the EU due to the coexistence in the continent of 19 sovereign bond markets. EU's fragmentation implies that each sovereign debt is associated with a different credit risk. Fanelli and Marsi argue this is particularly

relevant during debt crisis episodes. Indeed, although one may expect that an unconventional monetary shock would determine a positive correlation between the variations of risk premia on peripheral countries' bonds and risk-free rates, empirical evidence shows this is not always the case. In particular, periods of positive correlation alternate with periods of negative correlation between these variations, specifically clustered around debt crisis episodes. This puzzle can be explained through the spread shock. This shocks stems from the ECB taking fragmentation risk into account and playing an active role in the management of expectations on sovereign debt crisis. According to the researchers, a spread shock complements the monetary policy shock and a 'pure' information shock to reflect the idea that ECB's monetary policy does not affect the economy unidirectionally. Due to information asymmetry, it is rather a multidimensional and multifaceted object whose transmission does not exhaust into the categories of 'conventional' and 'unconventional' but crucially depends on information signaling, considered both in terms of ECB's general transmission of its policies to agents and in the form of its communication of policies to address fragmentation risk.

### 3.1.1 A factor model approach:

[Fanelli and Marsi(2022)] carry the identification of their *three* shocks through a novel methodology that combines different identification schemes: a factor model to extract factors that summarise EU market dynamics and a Proxy SVAR methodology to point-identify the three structural shocks of interest among the elements extracted from the set of admissible factors.

In the first step, a **factor model** is configured to extract a set of orthogonal factors, summarizing the dynamics of the Euro Area yield curve, sovereign spreads and the stock market. Letting  $\gamma_t$  be the 5x1 vector collecting HF surprises of OIS rates at maturities  $t = 1, 2, 5, 10$  years, respectively, and the STOXX50 index <sup>5</sup> The daily dynamics of  $\gamma_t$  are approximated by the factor model:

$$\gamma_t^{aug} = \Lambda \begin{matrix} f_t \\ 5x1 \end{matrix} + \begin{matrix} e_t \\ 5x1 \end{matrix} \quad t = 1, \dots, T \quad (2)$$

where  $f_t$  is a 3x1 vector of factors,  $\Lambda$  is the 5x3 matrix of time-invariant factor loadings and  $e_t$  is a vector of disturbances. For the two factors in  $f_t$  to be set-identified, a frequentist sign-restriction approach is applied. As it is known, for any 22 orthonormal matrix  $Q(QQ' = I, Q'Q = I)$ , the system above can be rotated and rewritten as:

$$\Gamma = (FQ)(\Lambda Q)' + e = F^*(\Lambda^*)' + e \quad (3)$$

where  $F^*$  and  $\Lambda^*$  are the rotated factors and loadings, respectively. The vector of rotated factors is  $f_t^* = Qf_t$ . The rotation matrices  $Q$ s are selected such that the model satisfies the following sign restrictions:

$$\begin{pmatrix} \gamma_t^{ag} \\ \Delta^{hf} OIS_{1y,t} \\ \Delta^{hf} OIS_{2y,t} \\ \Delta^{hf} OIS_{5y,t} \\ \Delta^{hf} OIS_{10y,t} \\ \Delta^{hf} (IT_{2y,t} - OIS_{2y,t}) \\ \Delta^{hf} (IT_{5y,t} - OIS_{5y,t}) \\ \Delta^{hf} (IT_{10y,t} - OIS_{10y,t}) \\ \Delta^{hf} STOXX50_t \end{pmatrix} = \begin{pmatrix} \Lambda^* \\ \begin{pmatrix} + & + & + \\ + & + & + \\ + & + & + \\ + & \cdot & + \\ + & - & - \\ + & - & - \\ + & - & - \\ - & + & + \end{pmatrix} \end{pmatrix} \begin{pmatrix} f_t^* \\ \begin{pmatrix} f_{mon,t}^* \\ f_{info,t}^* \\ f_{spread,t}^* \end{pmatrix} \end{pmatrix} \begin{pmatrix} e_t \\ \begin{pmatrix} e_{1,t} \\ e_{2,t} \\ e_{3,t} \\ e_{4,t} \\ e_{5,t} \\ e_{6,t} \\ e_{7,t} \\ e_{8,t} \end{pmatrix} \end{pmatrix} \quad (4)$$

where  $\Lambda^*$  is a matrix of rotated factor loadings and  $f_{spread,t}^*$  is the 'novel' spread factor that complements the monetary policy and the information factors,  $f_{mon,t}^*$  and  $f_{info,t}^*$  respectively. The signs associated with the spread

<sup>5</sup>As we are interested in unconventional monetary policies alone, these surprises are observed around ECB press conferences in the days that cover the period from January 2002 to June 2020. Besides, the time index  $t$  denotes days.

factor in  $\Lambda^*$  are partly informed by predictions formulated by Fanelli and Marsi in a stylised theoretical model they develop (more in Section 5 of [Fanelli and Marsi(2022)]).

In this phase, the sign restrictions are combined with other type of restrictions for the three factors to be set-identified. In particular, additional two narrative-type restrictions are employed.

These regard two empirical facts: 1) the fact that the spread factor tends to cluster around debt crisis periods or:

$$Var\{f_{spread,t}^* 1(t \in D_1)\} \geq \tau \times Var[f_{spread,t}^* (1 - 1(t_1 \in D_1))] \quad (5)$$

which maintains that the variance of the spread factor is  $\tau > 0$  times larger in the period 2010-2014 (debt crisis and in 2020 than in the other periods; 2) the fact that the impact of the spread factor on sovereign spreads must be relatively higher (in absolute value) than the impact that the information factor exerts on sovereign spreads. This motivates the symbol ‘-’ in 4. This magnitude restriction represents a natural characterization of the spread factor. Indeed, by nature, the spread factor must affect sovereign spreads (in terms of magnitude of the impact) more than any other factor extracted from HF surprises.

Finally, the factor model is estimated by *maximum likelihood* under the sign and the other type of constraints. This yields a set of 1000 factors  $f_t^* = (f_{mon,t}^*, f_{info,t}^*, f_{spread,t}^*)'$ ,  $t = 1, \dots, T$  denoted as  $\mathcal{F}_F$ .

In the second-step, the objective is to track the responses of financial variables of interest to the three structural shocks. To this end, three ‘representative’ factors are selected from the set identified in the previous step using [Fry and Pagan(2011)] methodology. These are denoted as  $f_t^{MT} = (f_{mon,t}^{MT}, f_{info,t}^{MT}, f_{spread,t}^{MT})'$ ,  $t = 1, \dots, T$ . Eventually, these three factors are used as external instruments in a (point-)identified proxy-SVAR estimated on daily data.

### 3.1.2 A Proxy SVAR methodology:

Given the three factors newly identified, we configure Fanelli and Marsi’s Proxy-SVAR model to jointly point-identify the dynamic causal effects produced by the **three** shocks on the target variables of interest.

To recover the structural form of vector autoregressive models (VARs, SVARs), one could recur to the traditional imposition of short- ([Sims(1980)]) or long-run restrictions ([Herwartz(2019)]). Further, to guarantee the economic content of the identified shocks, theory-based identification through sign restrictions [Canova et al.(2002)], [Faust(1998)], [Uhlig(2005)] could be employed. However, such approaches leave no room for the data to object against the structural model. The risk, that of intermingling model assumptions and conclusions. An alternative is presented by data-based identification techniques (i.e. identification through heteroskedasticity or independent component analysis (ICA); see [Gouriéroux et al.(2017)]; [Lanne et al.(2008)]; [Lanne et al.(2017)] Lanne, Meitz, et al.; [Rigobon(2003)]). These provide tools to test the otherwise just identifying structural model specifications. However, by its nature, data-based identification does not provide economic interpretation of resulting structural shocks. An approach that blends the benefits of theory and data-based identification schemes is the Proxy-SVAR. Indeed, on one hand the use of proxy variables requires economic foundation and theoretical assumptions that assure the orthogonality of the instrument to other structural shocks. On the other hand, the relevancy and validity of instruments can be tested and the structural model estimated under relatively weak assumptions. In Monetary Policy analysis, one could rely on the external evidence provided by [Romer and Romer(2004)] and [Gertler and Karadi(2015)].

Let’s first consider a finite-order VAR system:

$$Y(t) = \Pi_1 Y_{t-1} + \dots + \Pi_p Y_{t-p} + \mu + \delta t + u_t, \quad t = 1, \dots, T \quad (6)$$

$p$  is the number of lags, while the time index  $t$  represents days,  $Y_t$  is the vector of endogenous variables,  $\Pi(L)$  is a finite-order autoregressive polynomial,  $\mu$  is a constant,  $\delta$  are parameters associated with a linear trend that might characterize some of the variables in  $Y_t$ ; finally,  $u_t$  is the vector of reduced-form innovations assumed with time-

invariant covariance matrix,  $\Sigma_u$ .  $Y_t$  can be decomposed as  $Y_t = (r_t', x_t')'$ , where  $r_t$  contains the ‘policy indicators’, i.e. the variables that can be associated with the target structural shocks and  $x_t$  contains the response variables. The VAR innovations  $u_t$  can be decomposed likewise  $Y_t$ , i.e.  $u_t = (u_{r,t}', u_{x,t}')'$ . In a ‘standard’ SVAR context, the reduced form innovations are linked to the structural shocks by the expression:  $u_t = C\epsilon_t$ , where  $C$  is the matrix of coefficients that capture the *instantaneous* impact of the structural shocks  $\epsilon_t$  on the variables. While the reduced-form parameters in  $\Pi_i(L)$  ( $i = 1, \dots, p$ ) and the covariance matrix of residuals ( $\Sigma_u$ ) can be estimated consistently, the structural parameters collected in the non-singular matrix  $C$  are not identified without further theoretical and statistical assumptions. One possible solution to this identification problem is to restrict matrix  $C$  as the lower-triangular *Choleski* factor imposing a recursive causal structure among the model variables. This approach has a tradition in the identification of MP shocks, where the policy indicator is ordered at or near the bottom of the system [Bernanke and Mishkin(1992)]. However, it has been criticized that the Choleski scheme is inconsistent with a large class of DSGE models, which typically feature contemporaneous responses of macroeconomic variables to the policy shock [Castelnuovo(2012)]. As an alternative to exclusion restrictions - which have an interpretation of ‘internal’ instrumentation [Stock and Watson(2018)] - one can consider a measure of a structural shock of interest based on information **from outside** the VAR, and treat this measure as an external instrument for identification. First, a  $k \times 1$  vector of proxies  $z_t \equiv f_t^*$  is selected from the set of admissible factors  $\mathcal{F}_{t=1}^T$  obtained from the factor model. Second, given the *multiple shocks setup* ( $k > 1$ ), necessary additional restrictions (other than the instruments) are imposed on the model to obtain the IRFs. In particular, following [Angelini and Fanelli(2019)], we impose  $\frac{1}{2}k(k-1)$  restrictions for identification. Furthermore, in a Proxy-SVAR setup, only a subsets of the structural shocks in  $\epsilon_t$ ,  $\epsilon_{r,t}$ , is identified. Indeed,  $\epsilon_t = (\epsilon_{r,t}', \epsilon_{x,t}')$  where only  $\epsilon_{r,t}$  is instrumented, while  $\epsilon_{x,t}$  is the vector of non-instrumented structural shocks. Given this shock decomposition, the relation between the VAR innovations and the structural shocks becomes:

$$u_t = C_r \epsilon_{r,t} + C_x \epsilon_{x,t} \quad (7)$$

where  $C_r$  collects the on-impact responses of the variables to the target structural shocks  $\epsilon_{r,t}$ .

The **Impulse Response Functions** that track the impact of the target structural shocks on the variables are given by:

$$\frac{\delta y_{t+h}}{\delta \epsilon_{r,t}} = (SA^h S') C_r \quad , \quad h = 1, \dots, h_{max} \quad (8)$$

where  $A$  is the VAR companion matrix,  $S$  is the selection matrix such that  $SS' = I$ , and  $h_{max}$  is the maximum time horizon considered. The dynamic causal effects produced by the target shocks in  $\epsilon_{r,t}$  are recovered through the knowledge of the parameters in  $C_r$  and the reduced form VAR dynamics in  $\Pi(L)$  (compacted in the companion matrix  $A$ ). Let  $k$  be the number of structural shocks in  $\epsilon_{r,t}$ . In our instance, the structural shocks of interest are the monetary policy shock, the information shock and the spread shock, that is  $k = 3$  and  $\epsilon_{r,t}$  is a three-dimensional vector. The three structural shocks in  $\epsilon_{r,t}$  (i.e. the parameters in  $C_r$ ) are identified through a set of instrumental variables collected by the  $k \times 1$  vector of proxies (external variables),  $z_t$ . The instruments in  $z_t$  satisfy two conditions: the proxies  $z_t$  need to be relevant and exogenous (orthogonal) to the non-instrumented structural shocks,  $\epsilon_{x,t}$ .

1. The Relevance condition:

$$E(z_t \epsilon_{r,t}') = \Psi, \quad \text{rank}[\Psi] = k \quad (9)$$

where  $\Psi$  is the matrix of relevance parameters, connecting the *proxies* to the *target shocks* up to measurement errors. The condition maintains that the matrix  $\Psi$  must be full-column rank for the proxies carrying strong and independent information on the target structural shocks.

2. The Exogeneity condition:

$$E(z_t \epsilon_{x,t}') = 0 \quad (10)$$

The condition holds that the proxies  $z_t$  are orthogonal or exogenous to the non-instrumented structural shocks

in  $\epsilon_{x,t}$ .

The two conditions are summarized in the following equation:

$$z_t = \Psi \epsilon_{r,t} + \sigma_z \omega_t \quad (11)$$

where  $z_t$  is linked to the target structural shocks  $\epsilon_{r,t}$  by the matrix  $\Psi$ , up to measurement error term  $\sigma_z \omega_t$ .

The vector of MT factors  $z_t$  can be further specified as  $z_t = (f_{mon,t}^{MT}, f_{info,t}^{MT}, f_{spread,t}^{MT})$ , as a vector of three factors representatives of the monetary policy, information and spread factors in the set  $\mathcal{F}_{t=1}^T$ . Besides, to account for the *uncertainty* stemming from the selection of factors from the set  $\mathcal{F}_{t=1}^T$ , [Fanelli and Marsi(2022)] also estimate 1000 proxy-SVARs obtained in correspondence of all possible choices of instruments  $z_t \equiv f_t^* = (f_{mon,t}^*, f_{info,t}^*, f_{spread,t}^*)'$  in the admissible set  $\mathcal{F}_{t=1}^T$ . **The Augmented Constrained-SVAR (AC-SVAR):** By coupling the SVAR in Equation 6 with the auxiliary model for the external instruments of 11 the authors obtain the 'Augmented-constrained SVAR' theorized in [Angelini and Fanelli(2019)]. The advantage of their approach lies in the flexibility it provides. This enables the management of the additional restrictions necessary to point-identify the proxy-SVAR, other than the proxies. The system's configuration reads as follows:

$$\begin{pmatrix} Y_t \\ Z_t \end{pmatrix} = \begin{pmatrix} \Pi \\ 0 \end{pmatrix} S_{t-1} + \begin{pmatrix} \mu + \delta t \\ 0 \end{pmatrix} + \begin{pmatrix} u_t \\ Z_t \end{pmatrix} \quad (12)$$

$$\begin{pmatrix} u_t \\ Z_t \end{pmatrix} = \begin{pmatrix} C_r & C_z & 0 \\ \Psi & 0 & \sigma_\omega \end{pmatrix} \begin{pmatrix} \epsilon_{r,t} \\ \epsilon_{x,t} \\ \omega_t \end{pmatrix} \quad (13)$$

where  $\Pi = (\Pi_1, \Pi_2, \dots, \Pi_p)$  and  $S_{t-1} = (Y_{t-1}', Y_{t-2}', \dots, Y_{t-p}')$  and  $\epsilon_{r,t}$  is a  $k \times 1$ ,  $k > 1$  vector.

From this system, two moment conditions are derived:

$$\begin{aligned} \Sigma_{u,Z} &= \psi C_r' \\ \Omega &= \psi \psi' \end{aligned}$$

which marginalize out the parameters  $C_x$  that are not of interest as they capture the impact of the non-instrumented shocks on the variables.  $\Sigma_{Z,u} = E(Z_t u_t')$  is the reduced form covariance matrix that links the VAR innovations to the proxies, while the matrix  $\Omega = E(Z_t u_t')(E(u_t u_t'))^{-1} E(u_t Z_t') = \Sigma_{Z,u} \Sigma_u^{-1} \Sigma_{u,Z}$  is a nonlinear function of the reduced form covariance parameters in  $\Sigma_{Z,u}$  and  $\Sigma_u$ , respectively. The defined moment conditions connect the reduced form parameters in  $\Sigma_{u,Z}$  and  $\Omega$  to the structural (on-impact) and relevance parameters contained in  $C_r$  and  $\psi$ . For simplicity, the parameters in  $C_r$  and  $\psi$  are collected in the vector  $\theta$ . Estimation of the *structural* and *relevance* parameters collected in  $\theta$  is yielded as the solution of the following minimisation problem:

$$\min_{\theta} \left\{ \begin{pmatrix} \text{vec}(\hat{\Sigma}_{u,Z} - \psi C_r') \\ \text{vec}(\hat{\Omega} - \psi \psi') \end{pmatrix}' W_T \begin{pmatrix} \text{vec}(\hat{\Sigma}_{u,Z} - \psi C_r') \\ \text{vec}(\hat{\Omega} - \psi \psi') \end{pmatrix} \right\} \quad (14)$$

where  $W_T$  is a weighting matrix. The minimisation problem in 14 delivers a *Classical Minimum Distance* (CMD) estimator of  $\theta$  and it is carried out under  $l \geq \frac{1}{2}k(k-1)$  additional point-restrictions on the parameters in  $C_r$  and  $\psi$  under the multiple shocks scenario,  $k > 1$ .

**Remarks:** The advantage of framing a Proxy-SVAR estimation problem to recover the dynamic causal effects produced by the target shocks is that the  $l \geq \frac{1}{2}k(k-1)$  additional *identification restrictions* can be placed on the on-impact coefficients in  $C_r$  alone, on the relevance parameters in  $\psi$  alone. Alternatively, they can be distributed across the parameters in  $C_r$  and  $\psi$ , respectively. In our framework, we have  $k = 3$  structural shocks of interest and

thus  $l = 4 > \frac{1}{2}k(k-1) = 3$  additional restrictions. Following [Fanelli and Marsi(2022)], we impose three of them on the matrix  $\psi$  that is thus taken triangular and one on the matrix  $C_r$ , where a zero instantaneous impact of the spread shock on the measure of inflation expectations is assumed. This implies our Proxy-SVAR has  $l - \frac{1}{2}k(k-1)$  testable *overidentification* restrictions.

## 3.2 Transmission:

Modern Dynamic Macroeconomics studies the propagation of structural shocks ([Ramey(2016)]). At the core of this impulse-propagation paradigm are impulse response functions, i.e. the dynamic response of a macro aggregate to a structural shock. Following [Sims(1980)], [Bernanke et al.(1986)], and [Blanchard et al.(1986)], Structural Vector Autoregression (SVAR) analysis remains the most popular empirical approach to impulse response estimation. As the general interest lies in responses to multiple shocks, the SVAR approach maintains its supremacy as it provides estimates of the full system of responses. Over the past decade, an alternative to the method has emerged, with the introduction of Local Projections method (LPs) by [Jordà(2005)].

### 3.2.1 The VAR and the LP approaches to Impulse Response Functions:

How should we choose between SVAR and LP estimators of impulse responses in practice? Little theory has been developed to guide us to choose which method is preferable. By conventional wisdom, LPs are more robust to model **misspecification**, while SVARs are more **efficient**. VAR estimation of IRFs is a time series exercise that requires characterizing the *entire* dynamic system under consideration. The residuals of VAR models are treated as structural stochastic innovations, that is as a combination of economically meaningful shocks. The ones of interest are identified using theory-based assumptions or through external instruments (like in our case study). Post identification of these shocks, the autoregressive coefficients of the model are employed to study the transmission of the exogenous disturbances over time. This makes the modeling choice in a VAR very important for a correct inference of a shock's dynamic effects. Local Projections shift this system perspective to one where the impulse response are *directly* estimated with *univariate methods*, and without reference to other parts of the system. By non-requiring the correct specification of the underlying multivariate dynamic system, local projection provide an added benefit. Indeed, if the model is misspecified, the iterated method is more prone to bias. Only if the model correctly captures the data generating process (DGP), the estimates provided by a time series are consistent. Besides, the chosen lag order, an insufficient information set considered, unmodelled moving average components, and non-linearities ([Braun et al.(1993)], [Schorfheide(2005)]) are all common sources of misspecification. These sources of misspecification affect the identification of disturbances, variance-covariance decomposition, and the derived impulse response functions (IRFs). Then, VARs produce IRFs by iterating up to the relevant horizon the coefficients of a one-step-ahead model. Thus, if the one-step-ahead VAR is misspecified, resulting misspecified errors are compounded at each horizon in the VAR estimated IRFs. On the other hand, Local Projections estimate impulse response functions from the coefficients of direct projections of variables onto their lags at the relevant horizon. This way, the non-parametric method is more flexible and overall preferable to deal with the issue of misspecification. Secondly, while estimation of impulse response functions from Local Projections is a **multi-step** forecast done at each point, estimation of IRFs through VAR is a forecast at an increasingly distant horizon. So, LPs provide an additional benefit, that of avoiding miscalculations of impulse responses in the long-run. Thirdly, the VAR iterated method produces more **efficient** parameters estimates than the LP direct one. This is because LPs are a univariate semi-parametric approach, and as such they cannot compete in mean-squared error terms with the specification of a traditional structural multivariate time series model. Indeed, the more restrictions one can place in describing the data, the more efficient the estimates, the smaller the mean squared forecast errors, and the broader the scope to experiment with policy variations within the model. On a side note, however, empirical evidence suggests that LPs are *semiparametrically efficient* if their order is allowed to grow with the sample. Finally, LPs are preferable when investigating **non-linearities**, as non-linearities can be investigated more easily in univariate rather than in

multivariate models. Overall, the theoretical appeal of LPs is counteracted by the large estimation **uncertainty** that surrounds the coefficients’ estimates. In practice, from a classical perspective, the choice between the two methods is one that boils down to a sharp trade-off between bias and estimation variance.

### 3.2.2 The mechanics of Local Projections:

Local projections by [Jordà(2005)] are a generalized, but also computationally simplified way of estimating impulse response functions. Rather than deriving the dynamic multipliers as in Equation 8, we estimate them directly:

$$Y_{t+h} = \Gamma_h Y_t + \sum_{j=1}^p A_j Y_{t-j} + \eta_{t+h} \quad (15)$$

where  $\Gamma_h$  is an  $n \times n$  matrix describing how each variable loads on its own lags and on those of other variables between point  $t$  and some point in the future  $t + h$ . If the output measure is ordered first and the “shocked” interest rate third, then the coefficient of interest will be the first-row, third-column element of  $\Gamma_h$ , defined  $\gamma_h$ . Performing this regression for different horizons  $h$  between 0 and  $H$  gives us a vector of length  $H + 1$  with many  $\gamma_h$ s. Stacked together, they describe the impulse response function.[Plagborg-Møller et al.(2021)] show that for  $h \leq p$ , this impulse response is the same as the one obtained from a VAR. For larger horizons, Local Projection IRFs are typically less persistent and less “stable” because of the fact that the model imposes no dynamic structure whatsoever. This can be thought of as both a drawback and an advantage. LP impulse responses look less “nice” than VAR IRFs, but they are more “honest” representation of how economic variables dynamically relate to one another, while VARs impose a lot of structure onto these dynamics.

### 3.2.3 Road from a standard LP to an IV LP approach:

Local Projections are merely a *more direct* way to estimate impulse response functions but the challenges in terms of identifying the shock itself still apply. Our interest lies in estimating the impact of the *monetary policy shock* on *inflation*. We could either measure the shock directly, or rely on an ‘external’ source of variation that provide quasi-experiments to identify the causal effect of the shock. In a direct estimation approach, we would use the 5-year sovereign AAA-rated yields <sup>6</sup>, *ecb5y*, as a relevant measure representative of the monetary policy shock. Indeed, the central bank does not only intervene in the economy by influencing the short-term side of the term structure through manipulation of the policy rate, as done under conventional monetary policy; the central bank also acts on the long-term side of the curve decreasing the duration risk and credit risk ([Altavilla et al.(2015)]). When the effects on financial markets are studied, expansionary policy in the euro area is consistently found to lower long-term yields, reduce spreads and raise equity prices (for example, [Beirne et al.(2011)] [Benati(2010)] on yields and spreads). Thus, we may find the use of our *ecb5y* variable suitable as a measure of the monetary policy shock. Then, whether the related financial market effects are transmitted through to the real economy is another question. One difficulty with such method of directly measuring shocks is that it captures only part of the shock, and it is prone to measurement error ([Stock and Watson(2018)]). Long term sovereign yields measured around press conference dates may capture part of the shock related to unconventional monetary policy announcements but not all of it; for example, not the part revealed in speeches by EA members. Based on this approach, we would rely on recursiveness assumptions like in a VAR and estimate the contemporaneous impact matrix  $S$  as usual as the orthogonalized variance-covariance matrix from the reduced-form VAR residuals. On the other hand, recursion to an instrumental variable approach may result in *improved robustness*. In the instrumental variable approach, a shock is identified *externally*, through an external instrument being found to measure the shock. In our instance, the *monetary* shock identified in the Proxy-SVAR framework is called to serve as an instrument for *ecb5y*. The constructed variable is correlated with the true (unobserved) shock and, is uncorrelated with other shocks -posed

<sup>6</sup>This is a measure constructed by the ECB by averaging yields from euro area sovereign rates with triple A rating.

the arguments we made around exogeneity are correct. That is, the constructed variable is not the shock, but is an instrument for the shock. Then,  $\beta_h$  can be estimated even more directly by:

$$Y_{t+h} = \gamma_h z_t + \sum_{j=1}^p A_j X_{t-j} + u_{t+h} \quad (16)$$

where  $z$  is the shock or instrument for the shock, and  $X$  is a vector of control variables. Besides, if the shock is identified well,  $X$  does not necessarily need to contain all the lagged variables of  $Y$ , as they merely act as control variables. Given the robustness of the IV-LP method *à la* [Stock and Watson(2018)], drawing on [Fanelli and Marsi(2022)], we decided to adopt it. In particular, we employ the identified monetary policy shock as instrument to estimate IRFs in an LP approach, performing the IV version of an iterated multistep forecast. In the next section, the mechanics of this IV-LP methodology will be explored further.

### 3.2.4 An Instrumental Variables Local Projection (IV-LP) approach:

By [Stock and Watson(2018)], the baseline approach to LP is:

$$y_{t+h} = c_h + \beta_h x_t + controls + \eta_{t+h} \quad (17)$$

Where  $y_t$  is the response variable of interest,  $c$  is a constant,  $x_t$  is a scalar that measures or approximates the shock at time  $t$  and "controls" contains all the variables with respective lags that we are controlling for. This for  $h = 0, 1, \dots, H$ . However, in this framework, bias could occur if  $x_t$  was *endogenous* i.e.  $E[x_t \eta_t] \neq 0$ . In such instances, IV-LP provide the additional benefit of *robustness*, thus being preferable to the basic LP methodology. We consider an instrument  $z_t$  for the observed variable  $\tilde{x}_t$  such that:

$$(i) E[\tilde{x}_t z_t] = \phi, \quad |\phi| >> 0 \quad (18)$$

$$(ii) E[z_t \eta_t] = 0 \quad (19)$$

$$(iii) E[z_t u_t] = 0 \quad (20)$$

Conditions IV-LP(i) and (ii) are conventional IV *relevance* and *exogeneity* conditions, while condition IV-LP (iii) arises because of the dynamics. The former strictly requires that  $z_t$  is not linearly predictable given past  $\tilde{x}$ 's or that  $z_t$  is unpredictable given past values of  $\tilde{x}$ . Since we directly estimate the impulse response in one coefficient, we can use the standard errors on  $\beta_h$  to obtain confidence intervals. The only caveat is that the  $us$  will be correlated across  $h$ . To fix this, we need to adjust them for heteroskedasticity and autocorrelation, employing *Heteroskedasticity and Autocorrelation-Consistent* (**HAC**) standard errors. On a side note, HAC standard errors bring both advantages and disadvantages to our estimation. They estimate IRF point-by-point instead of extrapolating the period-by-period response implied by coefficients into the too distant future; in a way, this is a more "honest" description of the economy's dynamics. It is also more robust to misspecification and is computationally as simple as it gets. On the other hand, the approach leads to frequently unstable/"wobbly" estimates. Furthermore, only the sample for which all data, including  $z$  are available, can be used. Then, in our instance of a  $H = 25$  projection horizon, we lose 25 additional observations. Hence, inference becomes more challenging and estimates are less precise.

Post specification of HAC standard errors, we apply a two-Stages-Least-Squares (2SLS), with  $z_t$  as an instrument for  $\tilde{x}_t$ . The equation we are interested in is the following:

$$y_{t+h} = c_h + \beta_h \tilde{x}_t + controls + \nu_{t+h} \quad (21)$$

where  $y_{t+h}$  is the log transformation of the HICP,  $\tilde{x}_t$  is the observed *ecb5y* rate, so an imperfect measure of the monetary shock that we instrument through the extracted factor  $Z_{mon,t}$ . As we are interested in estimating the

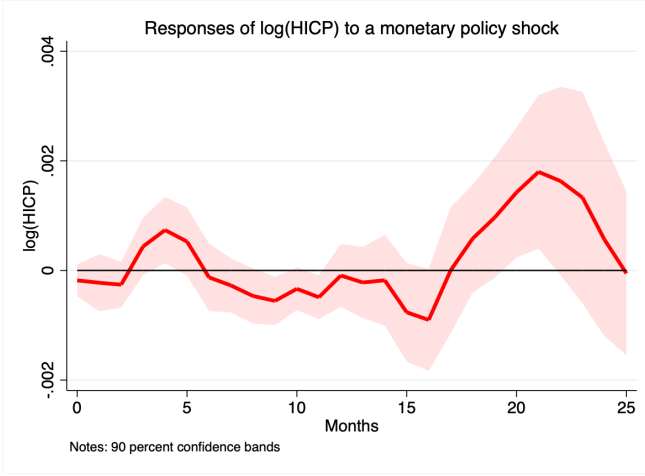
effect of a monetary policy shock on price level, we specify and then estimate our model in the following form:

$$y_{t+h} = \beta_{mon,h} e_{5y,t} + \underbrace{\alpha_h}_{constant} + \underbrace{\delta_h t}_{timetrend} + \underbrace{\psi_{1,h} f_{sp,t}^{MT} + \psi_{2,h} f_{info,t}^{MT} + \sum_{l=1}^p \pi'_{l,h} Y_{t-l} + \sum_{s=1}^q \phi_{l,h} Z_{t-s} + \gamma'_h d_t + \eta_{t+h}}_{controls} \quad (22)$$

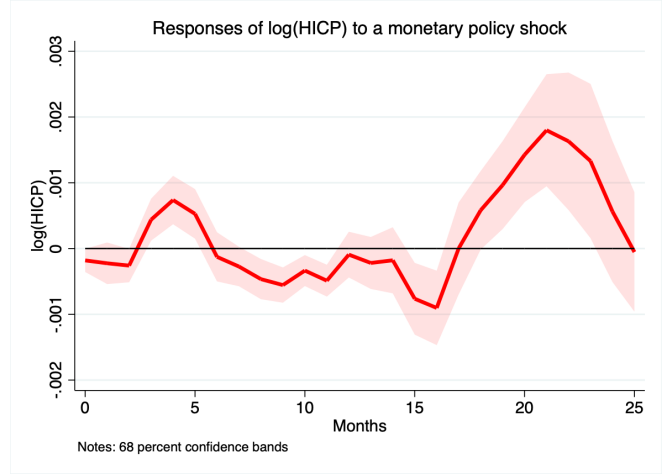
where we recall that  $y$  is the log of HICP and the max  $h$  is set to 25, i.e. we estimate the IRF at a response horizon of 25 periods. Moreover the number of lags for the control variables is set to 2.  $\alpha_h$  are constants,  $\gamma_h$  are coefficients associated with a linear trend,  $\beta_{mon,h}$  is the parameter that captures the impact of the policy indicator on the target variable after  $h$  periods;  $\eta_{j,t+h}$  are projection errors. The following controls are included in the IV-LP:  $p$  lagged values of  $y$ , and of other macro variables, in particular: the 5-year and the 1-year sovereign AAA-rated euro area yields, *ecb1y* and *ecb5y*, the 10-year spread IT-OIS, *spit*, the log of the STOXX50 index, *s50*, the log of the euro area industrial production index (excluding construction), *iprod*, the log of the HICP index, *hicp*, and the CISS index of systemic risk and financial distress produced by the ECB, *ciss*. Such control variables also include generic controls, such as the information and spread factors,  $f_{sp,t}^{MT}$  and  $f_{info,t}^{MT}$  (respectively), from our dynamic factor model; and  $q$  lags of the three proxies in  $z_t$ . Although LP-IV conditions in 18 and 19 hold, including control variables could reduce the variance of the regression error, and thus improve estimator efficiency. Indeed, control variables are functional to capture some of the dynamics of  $Y_{1,t}$ . As observed,  $\beta_{mon,j,h}$  picks out the impact of the policy indicator  $r_j$  on the variable  $y_{t+h}$  and is estimated by using the proxy  $z_{j,t}$  as an instrument for the policy indicator  $r_{j,t}$ . To sum up, we were interested in  $\beta_{mon,h}$  representing the effect of the monetary shock on the variable of interest. To estimate it we configured a regression in which the factor  $Z_{mon,t}$  extracted in the context of our Proxy-SVAR is used as an instrument. Besides, **Newey-West HAC** standard errors were employed for computation. Overall, local projections give a very flexible and simple way to compute **IRFs**, as don't require the specification of the underlying multivariate dynamic system they are more robust to misspecification compared to **SVAR** models. LP's are semi-parametric methods and as a consequence of this flexibility they are frequently unstable and wobbly showing higher variance and inefficiency of the estimator, differently from the SVAR methodology that if the model is correctly specified results to be more precise. In general iterative methods such as the VAR are more efficient, but even more prone to bias if the model is misspecified, while direct methods like LP suffer from higher estimation uncertainty. This of course arises the classical bias-variance trade-off.

## 4 Empirical Findings:

From figure (1) we observe the Impulse response of the logarithmic transformation of the harmonized index of consumer price (HICP) to a **contractionary** monetary policy shock. Besides, the IRF is estimated for 25 periods where each period  $h$  is a month. Furthermore, all target shocks have been normalised to decrease their respective policy indicators by 10 basis points. The empirical evidence reveals the response of inflation to monetary policy is **puzzling**. A few features emerge from the graphs. In the first quarter, prices rise unexpectedly as a result of a tightening in monetary policy. Only after the first quarter and then for a whole year, the log of the price index dampens quite consistently. At period 21, prices are expected to peak: an interest rate increase of 10 basis points caused by a contractionary monetary policy shock has an effect of roughly 0.0018 on the log of *hicp*, i.e. an increase of the 0.18% in the HICP (not logged). In other words, a interest rate increase of 1%, or 100 basis points, causes a sharp increase of the price causes a sharp increase of the price level of  $0.18\% * 10 = 1.8\%$  at period (month) 21. Then onwards, prices seem to revert to a downward pattern. Besides, estimation precision loosens as the horizon increases. In particular, we observe wider confidence intervals as  $h$  progresses. Furthermore, over the 25 periods we only find 5 statistical significant responses; the two that are negative are quite small compared to the positive ones, that as we said have important magnitude of positive sign and are statistically significant at the 90% level (see appendix B 6).



(a) IRF computed through LP-IV with monthly data, 90% confidence interval bands with Newey-West-HAC standard errors.



(b) IRF computed through LP-IV with monthly data, 68 % confidence interval bands with Newey-West-HAC standard errors.

Figure 1: LP-IV

We define the cumulative impulse response function for  $h$  periods as:

$$\sum_{h=0}^{H=25} \hat{\beta}_{mon,h} \quad (23)$$

This is the estimated cumulative effect in our 25 period window of the monetary policy shock on the price level. If we compute it by only considering statistical significant estimates (non-significant estimates are set to zero as there is not enough statistical evidence to reject the null  $\beta_{mon,h} = 0$ ) we obtain a cumulative effect of the monetary policy shock of roughly 0.02924 (0.002924 if normalized). By applying the same argument as before we can state that an increase of the policy rate of 1% caused a statistical significant effect on the price level of almost 3% in 25 months. Ergo, our empirical analysis carries a lot of uncertainty in the point estimates and does not fully support the deflationary effect of a contractionary monetary policy shock.

Nevertheless, before further conclusions are drawn, theoretical arguments are called to justify the dynamics seen in Figure (1). Our first reflection is that the response of inflation to a monetary policy shock is strictly related to the type of inflation that we are facing. Inflation could be **demand-driven** when it is mainly caused by an higher demand, or **supply-driven** if it is primarily due to a decrease in supply. It is common sense that the post-pandemic inflation surge has been driven both from the supply side, as the lockdown alongside with other dynamics has negatively affected the aggregate supply, and from the demand side, because of the expansionary monetary policy adopted by the ECB due to the pandemic response. What is relevant is that monetary policy has poor effectiveness in affecting supply-driven inflation. Macroeconomic theory<sup>7</sup> suggests that the central bank controls inflation by affecting the aggregate demand. This has been empirically documented in the EU by [Firat and Hao(2023)], who estimated the IRF to a monetary policy shock with respect to supply or demand driven inflation. Their research shows that supply driven inflation brings to a strong weakening in monetary policy transmission as monetary policy is more effective in shaping demand-driven inflation. Hence, to envisage the ability of the ECB to stabilize the post pandemic surge in inflation, we first need to assess and to take into account the supply and demand decomposition of inflation in order to provide a more detailed answer.

<sup>7</sup>see [Galí(2015)]

Moreover, we consider that ECB policies may not be effective in adjusting price levels if agents believe that achieving a declared target inflation (usually 2%) is not feasible. In this sense, the ECB and the general public different outlook of the economy could be responsible for the failure of unconventional monetary policy. In particular, the anchoring of expectations to the targets is a necessary condition for central banks to maintain price stability, as temporary shocks are prevented from having persistent effects on inflation [Neri et al.(2022)]. The features of our forecast in ?? call for a further investigation of the issue in the context of our research. If a central bank is fully credible, long-term expectations are insensitive to data releases [Bernanke et al.(2007)] and price patterns may truly speak about the efficiency of monetary policy. Without understanding the role of agents' expectations in the context of inflation stabilisation, conclusions on the effectiveness of monetary policy in stabilising inflation cannot be properly drawn. For the time being the issue hasn't been explored in the context of our research framework but this could certainly be interesting material for our next works.

Finally, in the last few years post pandemic, long-term inflation expectations in the euro area have been documented to have re-anchored to the 2 per cent symmetric inflation target of the ECB announced in July 2021 [Neri et al.(2022)]. Also, 2023 ended with an inflation rate of just below 3% in December, which is good news. Overall, the decisive actions undertaken by the ECB have helped keep long-term expectations well anchored so far. However, the risk of an upward de-anchoring of long-term inflation expectations represents an issue worth of close and continuous monitoring. Indeed, despite the gradual dissipation of the original shocks to energy and food prices that catapulted inflation above target, inflation is still high. It was 2.9 % in December 2023, up from 2.4 % in November 2023. Although prices have moderated recently, inflation has proven more persistent than expected. The recent competitiveness losses, owing to both exchange rate developments and energy price developments and the subsequent drop in exports suggest inflation still likely needs further intervention. In light of the previous arguments and of our empirical evidence, we conclude that the current outlook of inflation is surrounded by high uncertainty, as signalled by all types of expectations. Hence, as much as the tightening in monetary policy will contribute to inflation normalisation, additional efforts will likely be needed to achieve the ECB's 2021 objective.

## 5 Conclusions:

The capability of the ECB to lower the post pandemic surge in inflation is without any doubt a very complex and multidimensional question and the estimation of an IRF is an useful tool but not perfect one. Our empirical research follows [Fanelli and Marsi(2022)] for the identification of three monetary shocks in the EA and then [Jordà(2005)] for the LP-IV estimation of IRF of the monetary policy shock on the EA price level. Overall, we do not find any statistical evidence of the deflationary effect of a contractionary monetary policy shock. In addition, we observe that the effectiveness of a monetary policy shock is strongly related to the demand or supply driven nature of inflation and to the anchoring of inflation's expectations. Hence, given the extraordinary nature of the pandemic crisis with both supply and demand causes of inflation, we believe that this aspect, alongside with the possibility of a disanchoring of inflation's expectations, should be taken into account in assessing this question towards further conclusions.

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## 6 Appendix:

### SECTION A: DATA

| ID             | VARIABLE                     |
|----------------|------------------------------|
| <b>OIS1Y</b>   | 1 YEAR OVERNIGHT INDEX SWAP  |
| <b>OIS2Y</b>   | 2 YEAR OVERNIGHT INDEX SWAP  |
| <b>OIS5Y</b>   | 5 YEAR OVERNIGHT INDEX SWAP  |
| <b>OIS10Y</b>  | 10 YEAR OVERNIGHT INDEX SWAP |
| <b>STOXX50</b> | EURO STOXX 50                |
| <b>ITOIS2Y</b> | IT-OIS SPREAD AT 2 YEARS     |
| <b>ITOIS5Y</b> | IT-OIS SPREAD AT 5 YEARS     |
| <b>OIS10Y</b>  | IT-OIS SPREAD AT 10 YEARS    |

Table 1: HF Data from the Euro Area Monetary Policy event study Database (EA-MPD) observed around monetary policy announcements and used to extract the orthogonal factors.

| ID                   | VARIABLE                                                      |
|----------------------|---------------------------------------------------------------|
| <b>ECBYIELDAAA5Y</b> | 5-YEAR YIELDS AVERAGE OF AAA SOVEREIGN RATES IN THE EURO ZONE |
| <b>ECBYIELDAAA1Y</b> | 1-YEAR YIELDS AVERAGE OF AAA SOVEREIGN RATES IN THE EURO ZONE |
| <b>SPREADITOIS5Y</b> | IT-OIS SPREAD AT 5 YEARS                                      |
| <b>STOXX50LOG</b>    | LOG OF THE EURO STOXX 50                                      |
| <b>ILSWAP2Y</b>      | EUR 2 YEAR IRS INTEREST RATE SWAP                             |
| <b>USDEURLOG</b>     | LOG OF THE DOLLAR-EURO CHANGE                                 |

Table 2: Daily endogenous variables data.

| ID                   | VARIABLE                                                      |
|----------------------|---------------------------------------------------------------|
| <b>ECBYIELDAAA5Y</b> | 5-YEAR YIELDS AVERAGE OF AAA SOVEREIGN RATES IN THE EURO ZONE |
| <b>ECBYIELDAAA1Y</b> | 1-YEAR YIELDS AVERAGE OF AAA SOVEREIGN RATES IN THE EURO ZONE |
| <b>SPREADITOIS5Y</b> | IT-OIS SPREAD AT 5 YEARS                                      |
| <b>STOXX50LOG</b>    | LOG OF THE EURO STOXX 50                                      |
| <b>LOGINDPROD</b>    | LOG OF THE INDEX OF INDUSTRIAL PRODUCTION                     |
| <b>LOGHICP</b>       | LOG OF THE HARMONIZED INDEX OF CONSUMER PRICE(HICP)           |
| <b>LOGCISSINDEX</b>  | IT-OIS SPREAD AT 2 YEARS                                      |

Table 3: Monthly endogenous variables data, used for the estimation of the monetary policy shock effect trough the IV-LP.

## SECTION B: OUTPUT OF 2SLS REGRESSION

| h  | $\beta_{mon,h}$ | t-statistic |
|----|-----------------|-------------|
| 0  | -0.00180        | -1.03       |
| 1  | -0.00224        | -0.71       |
| 2  | -0.00259        | -1.02       |
| 3  | 0.00439         | 1.38        |
| 4  | 0.00738*        | 2.01        |
| 5  | 0.00527         | 1.40        |
| 6  | -0.00126        | -0.34       |
| 7  | -0.00274        | -0.91       |
| 8  | -0.00465        | -1.52       |
| 9  | -0.00555*       | -2.07       |
| 10 | -0.00335        | -1.43       |
| 11 | -0.00489*       | -2.02       |
| 12 | -0.000926       | -0.27       |
| 13 | -0.00219        | -0.56       |
| 14 | -0.00179        | -0.36       |
| 15 | -0.00763        | -1.40       |
| 16 | -0.00903        | -1.60       |
| 17 | 0.0000255       | 0.00        |
| 18 | 0.00581         | 0.97        |
| 19 | 0.00965         | 1.44        |
| 20 | 0.0143*         | 1.98        |
| 21 | 0.0180*         | 2.11        |
| 22 | 0.0163          | 1.56        |
| 23 | 0.0133          | 1.13        |
| 24 | 0.00563         | 0.53        |
| 25 | -0.000530       | -0.06       |

Table 4: Value of the estimated coefficients of the 2SLS regression with  $h = 0, \dots, 25$  with related t-statistic.

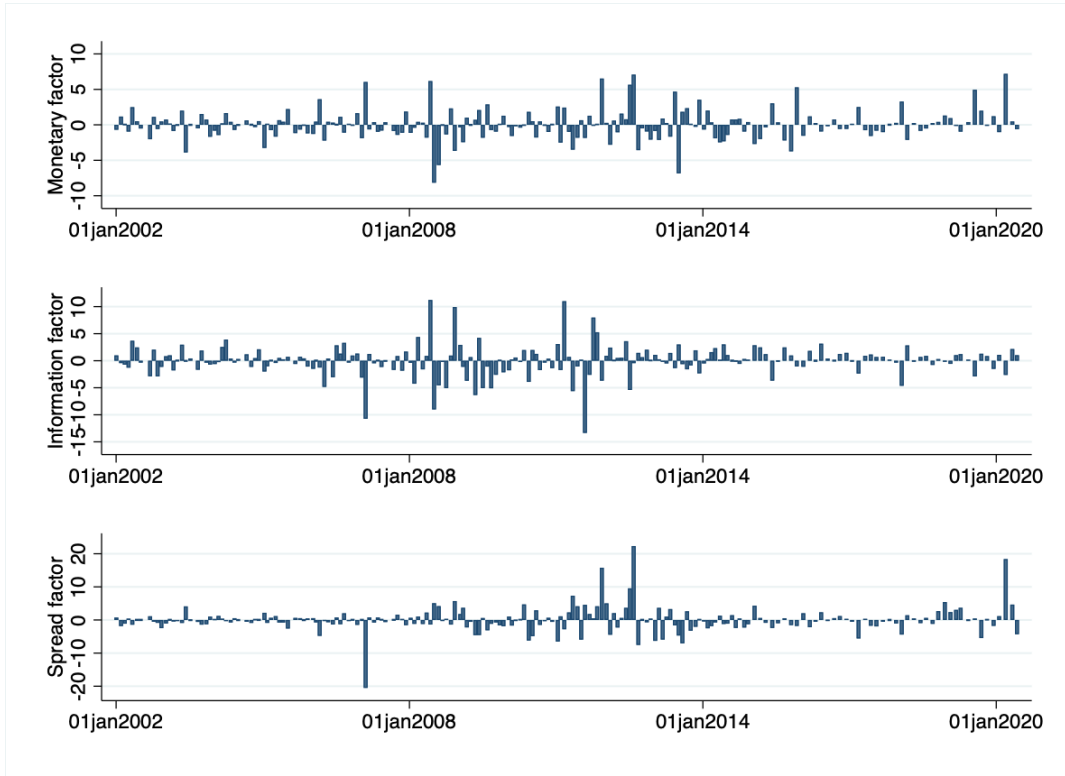


Figure 2: The three identified factors; the information and spread factor are normalized such that their loading on the 1-year OIS rate is equal to one, the same for the spread factor but the loading is on the 10-year spread IT-OIS. We have basis points on the y axis and time on the x axis